

Papermoon VR: Creating VR Landscape through Exploring VR Creation Methods

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ABSTRACT

Virtual reality technology offers creative practitioners to explore more ways of generating artworks as well as forms of art. VR head-mounted displays allow artists to create digital content with the freedom of experiencing the creations in real-time at flexible scales and positions. Through practice-led research, this project elaborates on the term VR creation, as a way of practice that uses VR native applications to create VR content. The project Papermoon was created using the VR application Quill by Smoothsteps and Oculus Quest 2, it was later built and presented using Unity. The paper shares insight into how artist practice with(in)VR through bodily movements. The paper also introduces aspects of the development of the artwork, particularly how it approaches the VR landscape, motivation, and techniques used for overcoming technical and aesthetic challenges.

Keywords: VR Landscape, VR Creation, Art, Bodily Gesture, Topology.

1 INTRODUCTION

In 2022, I was commissioned by Creative Brimbank, Brimbank City Council in Victoria, Australia, to create artworks and workshops that engage with its local community. The commission resulted in two outcomes: Papermoon, which is an outdoor projection behind the Brimbank City Council building entirely created and recorded using Virtual Reality (VR) native application Quill by Smoothsteps (hereafter Quill), and then Papermoon VR, an immersive artwork created using Quill and Unity, which was showcased at the Brimbank Library in July 2022.

The concept for the project first came to mind in 2021, after the COVID-19 pandemic had a profound effect on people's daily lives and emotional well-being. Over time, even if some people can walk outside for work and socialization, many remain constrained by social distance and mask requirements, or had been traumatized by the isolation experience of lockdown. As an artist, I wished to use creative mediums to give people companionship after sensing these unease in society. Papermoon often refers to the background props in theatre and photography for achieving an illusion of the environment. The artwork took inspiration from the term and uses the moon as a motif for conflict and continuity between worlds, a metaphor for solidarity and emotions. By experimenting with VR creation, I wished to conceive a papermoon that would always rise in the virtual landscape.

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Figure 1: VR Landscape overview.

2 THE LAND

During the early development of the artwork, I chose a local sightseeing place as a reference for the virtual landscape. The Kororoit Creek, located at the west of the Sunshine district, is a stream that crosses different communities and fosters many local floras and faunas. There are many stepping stones along the creek, which became small sightseeing places. The artwork took inspiration from Kororoit Creek's stepping stones and the plantations around it to create a VR environment based on artistic interpretations, as shown in Figure 1.

3 VR CREATION

Creative VR practices and methods have been rapidly evolving. This exploration is specifically interested in VR creation, a term that refers to both the action and outcome of creative practices using VR native applications. Like the word painting, it refers to both the generative action and the material outcome. As a landscape painter, this exploration was conducted in the hope of contributing to both technical and artistic insights into creating VR landscapes with(in) VR and for VR experiences.

VR creation is still a novel and strange practice whose parameters are radically different from my previous experience painting due to the intrinsic qualities of VR as a creative medium. Compared to other VR applications, Quill offers an organic experience that resonates with painting, where bodily movements drive formations of shapes, colors, and motions. On the other hand, VR sculpting and modelling frameworks might resonate more with sculptural artists and 3d modellers.

However, VR creation is fundamentally different from painting due to the intrinsic qualities of VR, for it is digital and post-

convergent [1]. Like other media forms, it is hard to be restricted to one sense, such as “sound art”, “visual art”, “digital art”, etc. [2]. VR creation foregrounds the use of body movements at a much larger scale, emphasizing the spatial relationship between the artists’ body and their creations. Meanwhile, the digital nature also emphasizes the temporal relationship of the body with the artwork, as the speed and force of movements can be recorded and replayed as digital tracings.

4 LAND-SKIN

The definition of landscape is constantly expanding. It first occurred in Dutch as “landshap” in the fifteenth century, meaning a piece of land. There is no doubt that the contemporary understanding and use of the term have long been more diversified [3]. VR (when using Quill) presents a virtual working space without borders and with a flexible scale, causing uncertainties in various regards, including the unfamiliar sensations of “floating” and questions such as coordinating gaze and composition. Probing these technical with artistic challenges and limitations leads to the final aesthetics and methods adopted in this artwork.

The surface of a landscape often consists of irregular, organic, and directional patterns. To compromise computing powers in manifesting natural topological structures, this work uses a method where digital pixels are arranged in different formations to create sheets of “skin”, each of these skins is designed to reflect specific characteristics of different materials surfaces (Figure 2). Through assembling, deformation, and changing the size of the pixels, one can fabricate large-scale terrains with effects of density and direction. Since any transformation is done through virtually grabbing and “painting” with the hand controllers, the specific actions are flexible based on how the whole-body coordinates. The tools are quite intuitive to use even for beginners as people often describe the act as “painting in air”. Nonetheless, it is worth our attention to scrutinize the intrinsic qualities of VR creation to explore the more creative and technical potential.

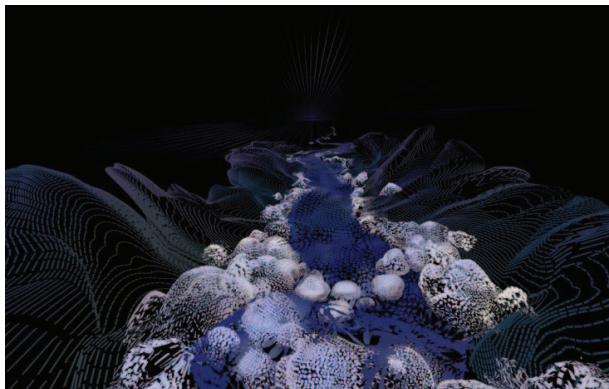


Figure 2: Close-up of the terrain.

5 THE RISING MOON

In contrast to telepresence, which offers access to a remote physical environment, VR offers a simulacrum, a synthetic structure [4]. Although the VR landscape of this artwork is in reference to a physical location, the thing itself does not represent any physical environment. Upon closer inspection, one can see that contours and motions here are all formed by basic rectangular

pixels that have been stretched in various ways. Like Caspar David Friedrich’s *Der Wanderer*, how the painting is made is crucial in conveying the feeling and atmosphere. The motivation of his paintings is pure, combining simple, restrained expressions of character (color, brushstrokes, etc.). For example, he would use a large number of extremely light dots to represent leaves and the first snow, witnessing the combination of two characteristics: delicacy and extreme concentration [5]. This VR landscape aims to draw on the way in which we see and relate to what is visible, to communicate the sensory and psychological aspects of the landscape, rather than just providing an understanding of reality. The flowing strokes and colors seek to explore the visible/invisible and inner/outer dynamics of things that (virtually) surround us.

The manifestation of the “papermoon” in this work calls for independence and possibilities of virtual environments. The wave of digital twins and mainstream creative approaches often point to the symbolic representations we experience in life, neglecting the creative potential of working with the unique qualities of VR. The rapid development of XR technology in recent years significantly impacted art forms and creative approaches, fostering more diverse experimentation of methods and aesthetics.

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AUTHOR’S BIO

Rosina is a New Zealand artist of Chinese descent who works across drawing, painting, digital mediums, and installation to explore modes of interactivity in both public and private spaces. Rosina moved to Melbourne, Australia in 2017 and completed her Bachelor of Fine Arts with Honours at RMIT University. She is continuing her artistic research as a Ph.D. candidate at RMIT University. Her research investigates different modes of spatial relationships through artistic practices of VR and public projections by working with local and natural environments. Rosina is actively involved in research and education programs focusing on digital technology and art. She has exhibited in China, Malaysia, Australia, and New Zealand.